

CSE 5830 Homework 2

9/16/2026

Due: 11:59PM, 9/23/2026

In this problem, you will implement a Naive Bayes classifier that can distinguish between spam and non-spam, or "ham" text messages. I recommend to use Python (Jupyter notebook or Google Colab) for implementation and submit your .ipynb file with the outputs (Please DO NOT clear the outputs).

For this assignment use the labeled text messages at:
<https://archive.ics.uci.edu/dataset/228/sms+spam+collection>

Download "smsspamcollection.zip" from the above website.

Split the data into a training set that will be used to train the classifier, and a test set that will be used to evaluate it. Use 80% for training and the other 20% for testing.

Do the following preprocessing for all text messages:

- Split the text by white spaces and punctuation marks
- Link the different forms of the same word (for example, "price" and "prices", "is" and "are") to each other
- Convert all words to lowercase so that the classifier does not treat "People", "people" and "PEOPLE" as three separate words.

You will write a program that has two parts: 1) learning, which learns a naive Bayes model from the labeled training text messages, and 2) classifying, which uses the model to classify the test data.

Training should include counting the frequency of each word in both spam and ham text messages and building a table that contains $P(\text{word} | \text{spam})$ and $P(\text{word} | \text{ham})$ for all words. Use the bag of words model. This is a pretty simple model which treats a piece of text as a bag of individual words, paying no attention to their ordering. For each word, calculate the percentage of times (frequency) it shows up in spam messages as well as ham messages.

For classification, if there is a word that is in the test data but is not in your training data ignore it.

Computation relies heavily on floating point multiplications. To avoid all the potential issues with multiplying very small numbers, perform a logarithm on the classifier equation to transform all the multiplications to addition.

Results:

Your results must be print out in your .ipynb file and also stored in a file called results.txt.

- Run your program when training on training data, then testing on the same training data.
- Run your program when training on training data, then testing on the test data.

Your program must print out its results in the following format (for both runs):

Size of ham training set: XXXX

Size of spam training set: XXXX
Percentage of ham classified correctly: XX.X
Percentage of spam classified correctly: XX.X
Total accuracy: XX.XX
False Positive: XX.XX

What to hand in:

submit your source codes as a .ipynb file with the outputs (Please DO NOT clear the outputs) and the results.txt file. Submit them through HuskyCT.

Notice:

Do not copy codes from the web. It is ok to search and learn how to implement, but then you need to write your own code.

You can use toolboxes for text manipulation. **But you need to write your own code for the Naïve classifier.**